

Scalability & Future Plan

Date	2 July 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Model retraining is a manual process	Delays incorporating newly released UNDP data	Medium
2	Batch upload is capped at 500 rows per request	Limits very large multi-country analyses	Low
3	Single-region deployment	Higher latency for users outside the hosting region	Low

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single server, ~50 concurrent users tested	Move to load-balanced containers on AWS ECS with auto-scaling
2	Data Storage	Single PostgreSQL instance	Migrate to managed RDS with read replicas
3	Performance	Synchronous model inference	Add a model-serving layer with caching (FastAPI + Redis) for faster repeated queries
4	Security	Basic JWT authentication	Add role-based access control (RBAC) and API rate limiting

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Automate UNDP / World Bank API sync	Q2 2026	Always up-to-date indicator data
Phase 2	Add multi-year trend prediction	Q3 2026	Enables forecasting a country's HDI trajectory
Phase 3	Mobile-friendly dashboard	Q4 2026	Wider access for field researchers
Phase 4	Public API for third-party integration	Q1 2027	Broader adoption by research institutions